



Illuminating
ENGINEERING SOCIETY

LIGHTING PRACTICE: DESIGNING QUALITY LIGHTING FOR PEOPLE IN OUTDOOR ENVIRONMENTS

AN AMERICAN NATIONAL STANDARD



ANSI/IES LP-2-20

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AN AMERICAN NATIONAL STANDARD**

Publication of this document
has been approved by IES.
Suggestions for revisions
should be directed to IES.

**Prepared by
The IES Lighting for Outdoor Public Spaces Committee**



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1.0 Introduction and Scope

1.1 Introduction

As the world's population continues to grow, a larger percentage of people are choosing to live in urban environments. Nowhere is this more apparent than in North America, where 82% of people live in urban areas.¹ In fact, the planet has gone through a process of rapid urbanization in recent decades, and in 2007, the global urban population exceeded the global rural population for the first time in history.

Outdoor urban space has become valuable property. Cities are being forced to implement large scale infrastructure projects to accommodate and plan for growing populations no longer constrained by buildings. People are more mobile, blurring the lines between work and social interaction. It is here, in the outdoor space, that people can stay connected with each other and the environment, enjoying a healthier and more social lifestyle, day and night.

New livable cities will attract people to walk, bike, and stay connected while using public or autonomous transportation. It is important that society acknowledge the need to design lighting for pedestrian based tasks. There also needs to be a reduction in acceptance of vehicular roadway-centric lighting as the only solution. Historically, the implicit lighting recommendations for outdoor areas are made from a vehicular vantage point, where pedestrians are considered a potential conflict. Vehicular lighting's role is for motorists to detect potential conflicts within the driving variables associated with viewing angle and moving at speed. Pedestrians and cyclists are more vulnerable than motorists; where they look and what they need in a lighted environment are completely different than for motorists. Intuitively, we know pedestrians need to detect sidewalk hazards for physical safety; see people, destinations, and attractions for navigation decisions; and recognize threats to achieve psychological confidence and reassurance.

Many of the techniques for interior lighting, such as the creation of multiple layers of light, can now be applied to outdoor spaces. What challenges does this present? This document will help lighting practitioners

understand ambient lighting levels, glare, contrast, and luminance to design for outdoor pedestrian spaces more holistically. It is important that outdoor public spaces at night provide sufficient light to engender the equivalent feelings of daytime reassurance, allowing people to enjoy public spaces. This document will guide users toward an understanding of how to design lighting for the pedestrian realm at night.

1.2 Scope

The charter of this Lighting Practice (LP) is to provide pedestrian-oriented lighting recommendations for the reassurance, safety, comfort, amenity, and enjoyment of pedestrians in outdoor environments. These recommendations provide a basis for lighting and space design, including the flexibility for application of multiple methods.

The common definition of *pedestrian* is one who travels on foot. For this LP, the inference is that pedestrians, traveling by foot, make decisions regarding their movements while moving at a pace slower than motorized vehicles. Therefore, the lighting recommendations contained within this document may be extended to any individual traveling in a manner consistent with someone on foot due to the absence of research for non-motorized transportation.

This LP makes recommendations beyond illuminance—which, considered alone, is inadequate as a means of accomplishing the complete visual experience necessary for pedestrian based tasks. Rather, it takes a comprehensive approach and makes recommendations based on zonal light level allowances, glare, adaptation, spectrum, and contrast while addressing safety, timing, and the need for reassurance. Application of these recommendations will ultimately enhance the pedestrian's visual experience while also respecting the environment.

A number of IES Recommended Practice (RP) documents provide recommendations and design guidelines for specific outdoor lighting applications. This document is not intended to supersede those publications. Instead, it supplements the various RPs, augmenting them in subject areas not otherwise addressed.

2.0 Fundamental Considerations for Pedestrian Lighting

2.1 What Pedestrians Need to See and Why

Nighttime lighting designed specifically for pedestrians (i.e., people walking outdoors) is often very different from the sidewalk light generated by adjacent buildings or vehicular lighting systems. Pedestrians have a priority of tasks after dark; therefore, they need a lighting system capable of benefiting multiple tasks, such as navigation and enjoyment of public spaces. For example, the lighting necessary for deciding the direction in which to navigate and then avoid hazards may be different from the lighting needed in areas designated for social interaction and enjoyment. What to illuminate and why are critical components of pedestrian lighting.

Whether for walking, cycling, resting, or socializing, pedestrian based lighting systems need to accommodate human nighttime vision, and the visual experience needs to be tailored toward helping pedestrians accomplish their tasks. Appropriate lighting design for nighttime pedestrians places a priority on how people see and the tasks they need to perform. These considerations have the ability to increase pedestrian reassurance, activate outdoor spaces, and begin the positive cycle of attracting more pedestrians to outdoor public spaces at night.

2.1.1 Orientation and Wayfinding. In the hierarchy of nighttime pedestrian tasks, the most important are orientation and wayfinding. Orientation—the ability to comprehend one's surroundings, other pedestrians, and possible threats—and wayfinding—the ability to make movement decisions and navigate a directional path—are the prerequisites for pedestrian reassurance and terrain safety.

2.1.2 Reassurance. Second in hierarchy is line-of-sight detection of possible threats, including those from other pedestrians, and detection of surrounding boundaries and egress points. The identification and ranking of threats require information to be recognized as early as possible so that there is an appropriate amount of time to make decisions and react. It should be said that if a space is unsafe during the day, lighting conditions at night will not make it safer.

Reassurance may be defined as that which provides comfort to make someone feel less worried, afraid, or doubtful, and which restores confidence. Good lighting can reduce the degree to which reassurance deteriorates after dark. Reassurance is something that can be influenced, and it is critical to pedestrians' willingness to engage with their surroundings and ultimately activate a space along with other pedestrians.

2.1.3 Pedestrian Safety. Third in hierarchy is the physical safety of pedestrians against hazards caused by terrain navigation and moving objects. For safe movement at night, pedestrians need the ability to discern expected terrain changes such as stairs, but they also need to identify unexpected obstacles, including:

- Changes in pavement or grade, such as raised sidewalk segments or sinking potholes
- Curbs and wheel stops
- Slippery conditions
- Moving pedestrians, skateboarders, and bicyclists

2.1.4 Atmosphere and Enjoyment. After the basic requirements of outdoor pedestrian lighting are met, supplemental and decorative lighting may be considered. Outdoor spaces often include furniture designed for specific activities such as checking a phone, socializing, resting, and visual enjoyment. These different activities and settings may require application specific lighting. For pleasing spaces, views of architecture, vertical and horizontal surfaces, people, and nature need to be included.

2.2 Light and Human Health

Research concerning non-visual human responses to optical radiation (light) has created new and serious discussions about the effects of exterior lighting on human health. Since the effects of lighting can be profound for human health and well-being, it is increasingly important for lighting practitioners to understand the biological effects of light/dark cycles.

Lighting does not just affect our visual sensation and ability to see; it is also the primary stimulus for the biological timing of many physiological and neurobehavioral effects. In fact, there are myriad ways lighting impacts human health. Perhaps, the most

important to understand is its ability to disrupt the internal circadian body clock. There are predictable environmental changes and rhythms to which humans have evolved with respect to daily (day/night) and seasonal cycles. Nearly all physiological and behavioral parameters exhibit circadian rhythms. As a result, human events including the sleep/wake cycle, hormone levels, core body temperature, and blood pressure can be affected, and synchronization of the circadian clock is paramount to efficient function and increased chances for good health and survival.

Five key factors within the lighted environment drive our circadian system: the *quantity* of light at the retina, the *duration* of exposure to that light, the *spectrum* of the light, the *distribution of the light* in the observer's field of view, and the *timing* of the light exposure relative to the circadian clock.

Research into the impact of light on human health is in its early stages and characteristics of light required to impact the circadian clock are as yet not completely known. Preliminary data suggests light levels as low as 4 to 5 lux, measured at the cornea, could affect circadian rhythms but this also depends on the timing and duration of exposure. There is, however, a greater body of research regarding how spectrum can affect the circadian clock.

This LP supports and incorporates the latest information regarding optical radiation and human health to make lighting recommendations for outdoor pedestrian environments. Lighting practitioners should work with communities to determine appropriate light levels, light source spectrum, and luminaire light distribution characteristics to meet a community's preference and task requirements. Using a thoughtful application-oriented design approach guarantees that light shines where it is needed and when it is needed, limiting negative impacts on human health, and the light/dark cycle especially. Pedestrian environments built for activity and enjoyment will tolerate lighting parameters different from those used for navigation. Stray light will always be considered unwanted and therefore a potential concern.

As with any research concerning human health, the information is complex and often changing. The reader is encouraged to refer to *IES TM 18-18, Light and Human*

Health: An Overview of the Impact of Optical Radiation on Visual, Circadian, Neuroendocrine, and Neurobehavioral Responses, and to search for current information on this subject.²

2.3 Minimizing Environmental Impact

While electric illumination of the industrialized world has been instrumental in the creation of nighttime commerce, it has also prevented much of today's world population from experiencing the night sky, a historically crucial orientation and source of knowledge about the world. Urban residents in particular are deprived of the opportunity to observe the star-filled night sky, while some have warned, "the cultural loss of this development... can hardly be overestimated."³

The culprit? Light pollution, a broad term that encompasses all the adverse or obtrusive effects of electric light not under control, which negatively impacts the environment through stray light, light trespass, and sky glow.

Pedestrian-based systems that over-light are a major concern addressed in this LP, which will also identify outdoor lighting design considerations related to the contribution of electric lighting systems to environmental light pollution. Additional information on the environmental impact of light pollution may be found in *ANSI/IES LP-11-20, Lighting Practice: Environmental Considerations for Outdoor Lighting*.⁴

2.3.1 Sky Glow. This form of light pollution is easily recognized for its negative impact to the environment. Artificial (or human-made) sky glow* is not just due to ill-advised or uncontrolled stray light pointing upward into the night sky, but rather a combination of uplight, over-lighting, and increased use of short-wavelength spectrum. Whether intentional or unintentional, any light emitted or reflected above horizontal, including that reflected from the ground or vertical surfaces, contributes directly to sky glow.

* The brightening of the night sky that results from the scattering and reflection of light from the constituents of the atmosphere (gaseous molecules and aerosols), in the direction of the observer. It has two separate component: natural sky glow and artificial sky glow. – Ref. *ANSI/IES LS-1-20, Nomenclature and Definitions of Illuminating Engineering*. Online: www.ies.org/standards/definitions/. (Accessed 2019 June 21).

Both the types of luminaires used and the design of an outdoor lighting system can contribute to the damaging impact of sky glow. Design solutions helping to minimize sky glow include: directing light from exterior lighting systems below 80 degrees vertical; limiting light that does not illuminate the targeted area; not over-lighting; and dimming all lights after the specified time of use unless necessary for safety and security. As more light can be economically produced with energy-efficient LEDs, it is important that lighting practitioners exercise prudence to use proper illumination levels and no more.

The trend toward using LEDs has created a subsequent increase in short-wavelength (blue-rich) light sources on streets and in other outdoor spaces. Recently developed analysis tools are capable of comparing different spectral power distributions (SPDs) to demonstrate how spectrum can affect sky glow.⁵ This tool can enable lighting practitioners to experiment and compare the uplift and spectral impact between different lighting systems. (For additional information on light pollution and sky glow, refer to ANSI/IES LP-11-20.⁴)

2.3.2 Light Trespass. This is a form of light pollution that includes unwanted stray light on surrounding properties. The design methods discussed for reducing sky glow—minimizing emissions that miss a major percentage of their target, keeping lights off when not needed, and not over-lighting—may also help limit light trespass, sometimes described as the “light shining in my window.” Further corrections might include shielding offending luminaires, selecting luminaires with tightly controlled distributions, and careful consideration of mounting location and height of luminaires in the context of the surrounding area. (For additional information on light trespass, refer to ANSI/IES LP-11-20.⁴)

2.3.3 Spectrum. Any electric light that damages the flora and fauna of the nighttime environment is a concern. Light pollution and light spectrum each represent areas of environmental concern. Certain zones, for example turtle nesting areas, may need special consideration when it comes to limiting unwanted light, limiting the spectral content of the light, and preserving dark adaptation. Of course, all lighting is situational, and solutions should seek to reflect a balance between

curbing light pollution and providing for the relevant visual tasks. (For additional information on spectrum as it relates to the environment, refer to ANSI/IES LP-11-20.⁴)

2.4 Exterior Lighting Zones

Zoning is a well-established practice in community planning. The fundamental idea behind zoning is that it allows a community to determine and regulate appropriate use of different areas.

Introduced by the International Commission on Illumination (CIE) in 1997 and adopted by the IES in 1999, lighting zones for the exterior environment were originally implemented to help reduce light pollution. In the years since, lighting zones have become a useful tool to design environmentally responsible solutions and to enable quality of light decisions based on the prescribed ambient lighting conditions of adjacent properties. Lighting zones will be referenced throughout this document as one of the primary influencers of lighting design and light level decisions.

Lighting zones give communities the opportunity to set lighting limits, thereby establishing a predictable amount of ambient lighting within certain areas. (See **Section 4.1.1** for more information on community goals and themes.) Lighting zones help minimize the contrast (and conflict) between extremes in lighting such as a brightly lighted car dealership or sports facility adjacent to, or within line of sight of, a residential neighborhood. When used properly, lighting zones facilitate minimal changes in visual adaptation when traveling between areas, and allow practitioners the flexibility to use different lighting techniques without the burden of excessive regulation.

The choice of an appropriate lighting zone is a matter of judgement based on community priorities for any given area. It is recommended that the lowest reasonable lighting zone(s) be adopted. Selection of lighting zones should be based not on existing conditions, but rather on the type of lighting environments the jurisdiction seeks to achieve.

Additional information on lighting zones including definitions and examples, may be found in ANSI/IES LP-11-20.⁴

2.4.1 Lighting Zones. As used in the *Joint IDA-IES Model Lighting Ordinance (MLO)*⁶ and defined in ANSI/IES LP-11-20, lighting zones are summarized here, along with suggested uses.

LZ-0, no ambient light: applies to areas where the natural environment could be seriously and adversely affected by even small amounts of electric lighting at night. Human activity here is sparse and subordinate in importance to the natural environment. While some lighting is allowed, it is required to be controlled. The vision of human residents and users is adapted to the very low natural light levels, and they do not expect to see electric lighting.

LZ-1, low ambient light: applies to developed areas within a natural environment and areas of human activity that are inherently dark at night. Electric lighting at night could adversely affect the quiet, dark character of the area. The vision of residents and users is adapted to the low light levels, and they do not expect to see electric lighting except where it is needed to improve visibility and safety. Low light levels are appropriate and expected to be non-continuous (i.e., pools of light rather than uniform lighting along a path or roadway).

LZ-2, moderate ambient light: applies to areas of human activity (i.e., habitation, recreation, and/or work) where electric lighting may be required for safety and convenience at night. The vision of human residents and users is adapted to moderate light levels, and they have moderate expectations of electric lighting. Lighting is expected to be non-continuous (e.g., pools of light at intersections rather than uniform lighting along a path).

LZ-3, moderately high ambient light: applies to areas of human activity (i.e., habitation, recreation, and/or work) where electric lighting may be continuous and is required for safety and convenience at night. The vision of human residents and users is adapted to moderately high light levels, and they have moderate to high expectation of electric lighting. Lighting is expected to be continuous (e.g., lighting delivery fairly even along the length of a path or street).

LZ-4, high ambient light: applies to areas with high levels of human activity at night, including significant interaction among pedestrians and/or vehicles. The

vision of humans when outdoors in these areas is typically adapted to moderate light levels. Lighting is continuous and is required for safety and convenience. Expectations for electric lighting are high, both in terms of light levels and uniformity along pathways or streets.

3.0 Considerations for Pedestrian Vision

The subsections that follow focus on the predominate conditions of outdoor built environments as described in lighting zones LZ-1 through LZ-3. It is in these environments, research shows that most urban and suburban pedestrians experience a high-mesopic (1 to 5 candelas per square meter* or low-photopic (5 to 10 cd/m²) adaptation.⁷ This is significant because the physiology of the human eye in these conditions leverages different receptors that are particularly efficient at different tasks. Particularly at mesopic light levels, the rod receptors are extremely useful in peripheral vision, key to accomplishing the tasks of a nighttime pedestrian.

3.1 Glare

As lighting designers will tell you, a primary objective to creating a great visual environment, and a commonly found trait in many of their favorite projects, is for users to admire a project without consideration for where light is coming from. Why is this? Is it to spare the observer from seeing hardware and luminaires when they should be focused on architecture or landscape? Or is it to create the best visual experience by eliminating glare from the observer's field of view? Either endeavor by itself would be worthy, but mitigating glare from the field of view is the act of higher purpose, as it can be fundamental for achieving the prescribed visual design criteria for the project.

Glare is caused when bright light enters the eye, either directly from a light source (direct glare) or indirectly due to reflection off a surface (reflected glare). In addition to glare type, the magnitude of glare can also vary. In the

* Pavement illuminance (lux) may be determined from luminance: Multiply by π and divide by the pavement reflectance. *Note:* Assumes a Lambertian surface.

context of glare, *bright light* means a source that is so much brighter than the surrounding environment it can cause discomfort and/or visual impairment, the latter commonly named *disability*. Although there may be separate physiological mechanisms for the discomfort and disability glare responses, they can occur at the same time.

Discomfort glare can be uncomfortable, distracting, or even painful, and can cause an aversion response even if it does not reduce visibility or affect adaptation.

Disability glare is caused primarily by bright sources close to the axis of view that are much higher in luminance than the viewer's adaptation level. Disability glare inhibits vision because it causes a photochemical or localized scatter event that casts a luminous haze upon the retina. Unable to accommodate the scatter and resolve the haze, our visual system works to shift into a higher adaptation level. The unwanted result of a higher adaptation level at night is an increased difficulty in resolving excessive bright-to-dark ratios, meaning a potential reduction in the perception of contrast. This reduction in contrast is an issue of task visibility rather than visual comfort. (See **Figure 3-1** for an example of glare.) (Note: Age-related changes in the eye result

in older observers being more prone to scatter in the ocular media and the disabling effects of glare.)

For reasons discussed in this section, the principles of glare reduction should be considered along with illuminance in lighting designs for exterior areas. Just as pedestrians during the day can use sunglasses to reduce brightness and glare, making eyes more comfortable, and drivers can use visors during the day and use dimmable rear-view mirrors at night to avoid becoming visually disabled by glare, this LP aims to leverage available glare metrics to complement illumination recommendations.

With several metrics, theories, and points of view available, glare remains a topic of ongoing research. **Annex B** provides additional discussion regarding glare, including formulas for calculating glare. One limitation presented in **Annex B** describes how the widely accepted Unified Glare Rating (UGR) does not accommodate the high luminaire position angles found in many pedestrian lighting systems (i.e., higher than 55° above the axis of view). For this reason, this LP uses the Glare part of the Back light, Uplight, Glare (BUG) rating system defined in *ANSI/IES TM-15-20, Luminaire Classification System for Outdoor Luminaires* (LCS) in its recommendations pertaining to the mitigation of glare.⁸



Figure 3-1. Example of how glare can negatively affect pedestrian vision at night, making it difficult to resolve details within bright and dark areas at the same time. (Image courtesy of Rick Utting)

The LCS defines the distribution of light emitted from a single luminaire within ten defined zones, shown in image **Figure 3-2**. The LCS data comes from a manufacturer's photometric test report of a luminaire and becomes the basis for the IES BUG rating system. The lumens emitted from a luminaire that fall within four of these zones (FVH, FH, BVH, BH) are compared to pre-determined thresholds to define the Glare (G) rating of the luminaire. For lighting equipment mounted above eye height, it is often the light found in the range of 60° to 90° above nadir that enters a pedestrian's field of view against the darkness of night. The BUG ratings help determine the appropriateness of a luminaire for pedestrian use, thereby aiding lighting designers in making equipment decisions for people in outdoor environments.

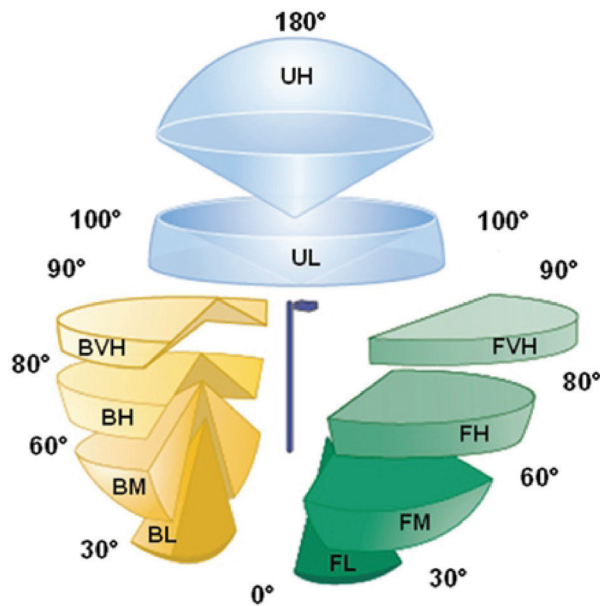


Figure 3-2. The breakdown of the Luminaire Classification System's zones, used in determining a luminaire's BUG rating. (Image © Illuminating Engineering Society)

It only takes a single light source to create glare. In fact, a bright luminance within the field of view can produce glare and inappropriate adaptation. Design techniques have been known to reduce glare include:

- Lumen output reduced to provide appropriate illuminance (as specified or via dimming)
- Shielding the direct line-of-sight (regressed light source, solid shield, louver, or baffle)
- Diffusion of the light source (translucent filters, screens, or fabrics)

- Use of reflected light or edge lighting
- Use of less specular and more matte or textured surface materials

3.2 Light Levels

It is important that lighting designers consider available lighting metrics such as glare, adaptation, and uniformity when designing exterior lighting systems. If done correctly, it may be possible to reduce overall light levels and simultaneously improve vision.

Illuminance, measured in lux, describes the amount of light incident upon a surface, such as a stairway or pedestrian walkway. However, this is not what the eye sees. The eye sees reflected light from a surface (quantified as *luminance*) rather than the light incident upon that surface (*illuminance*). It stands to reason that luminance-based design would be ideal; however, variations in surface reflectances and the direction of pedestrian gaze make luminance an impractical metric when designing lighting for people in outdoor environments. This LP describes and explains how reflectances are observed, thus allowing luminance to be a consideration in lighting design decisions. In reality, a good portion of lighting design is done without the ability to select or maintain colors and materials. For this reason, the amount of light needed in a pedestrian environment is most often specified using illuminance.

Historical design guides tend to specify horizontal illuminances only, like those on the pedestrian walking surface. Since vertical surfaces such as people's faces and building façades are also objects of pedestrians' attention, there is benefit in specifying vertical surface illuminance. The broad spatial distribution of conventional lighting (mainly high pressure sodium and metal halide) traditionally allowed designers to reasonably assume that there would be sufficient light on vertical surfaces. This assumption may no longer be valid following the introduction of LED light sources using well-defined optics and putting light only where needed.

Illuminance is an important consideration because it affects the ability of pedestrians to perform visual tasks. Appropriate illuminances can enhance the ability to detect pavement surface trip hazards⁹ and the degree to which the intentions of other people can be evaluated.¹⁰

Appropriate levels of ambient light can also encourage pedestrian reassurance and activity.¹¹⁻¹³ While there is a general trend to believe higher illuminances can improve these tasks, it is important to understand that there are caveats associated with increased light levels. First, research consistently shows a threshold at which further increases in illuminance bring negligible visual gains.¹¹ Second, increases in illuminance for visual benefit need to be considered alongside the associated disservice to the nighttime environment, energy costs, and degradation of the natural world.

Section 4.0 provides additional information on what to illuminate and how to determine appropriate light levels in the outdoor environment.

3.3 Adaptation to the Visual Environment

Adaptation to the visual environment is possible because of the eye's mechanical, chemical, and neural ability to change. Incredibly, the human eye is able to facilitate vision within a very wide range of light levels, from moonlight (approximately 0.1 lux) to bright sunlight (approximately 100,000 lux), but not all at the same time.

At night, color perception is reduced, the adaptation threshold is lower, and there is a lower maximum-to-minimum luminance range, about 1,000 to 1. Luminances higher than the adaptation range will be perceived as glare, and any luminances lower than the range will appear as black, with no details visible.¹⁴

An example of the adaptation process is found in watching a matinee movie. Upon exiting the theater, the daylight is blinding, but after a few minutes it is once again comfortable. Conversely, upon entering a dark theater, details such as steps and empty seats are difficult to discern, but after several minutes the adapted eyes can clearly see the surroundings.

Understanding this condition, designers should consider a project's lighting zone and provide for gradual transitions to facilitate the eye's adaptation periods. (See **Sections 2.4, 4.1.1, and 4.1.2** for lighting zone definitions, community planning, and transitions, respectively.) For example, pedestrians leaving a brightly illuminated sports field (approximately 1,000 lux) might need to transition to an adjacent parking lot (typically about 5 to

10 lux). To facilitate adaptation, the lighting professional should consider designing for a gradual decrease in light level along the pathways leading to the parking lot.

The adaptation process takes time. Typical adaptation times will vary depending upon the luminance disparity and the observer's age, ranging from a few seconds up to 15 minutes. Furthermore, it is more difficult to go from brighter light levels down to low light levels than the reverse, and older eyes adapt more slowly. (See **Section 4.1.2** for more detail regarding transitions between light levels.)

The concept of adaptation explains why it is important to control glare from high-brightness luminaires: important visual details may be lost above or below the adaptation range. (See **Section 3.1** for more detail regarding glare.) Even at low light levels, luminaires that emit light in the direction of a pedestrian can create brightness perceived as glare, reducing the ability to see important visual details. For this reason, caution is required when selecting luminaires—especially step-lights, bollards, and pole mounted luminaires—that are excessively bright or that emit light at angles close to horizontal.

3.4 Uniformity

The illuminance across a lighted area is not equal at all locations; it varies, reaching a maximum in the vicinity of a light source and a minimum mid-way between two successive light sources or somewhere near the perimeter (see **Figure 3-3**, for example).



Figure 3-3. The term *uniformity* refers to the variation in illuminance variation across a lighted area. Shown here is a path with soft transitions between high and low light levels along the path. (Image courtesy of Landscape Forms)

When designing a lighting system or validating an existing one, it is standard practice to calculate the illuminance at each of a series of points (nodes) across the lighted area (see **Figure 3-4**). The arithmetic mean of these illuminance values is used to represent the average light level. The uniformity, often expressed as the ratio of average illuminance to minimum illuminance, can then be calculated:

$$\text{uniformity} = \frac{(\text{average illuminance})}{(\text{minimum illuminance})}$$

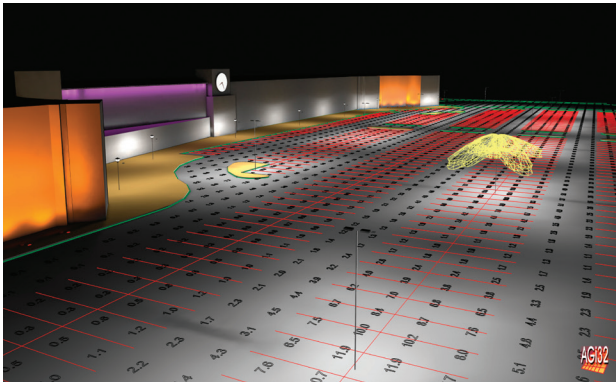


Figure 3-4. Lighting design software can predict illuminance values, including uniformity. (Image courtesy of Lighting Analysts, Inc.)

One situation where the uniformity may be critical is pedestrian reassurance because it characterizes the relative brightness and darkness within the same field of view. Average illuminance alone says nothing about the spatial variation in illuminance. The same average illuminance is possible from installations with high or low uniformity, and those with low uniformity can consist of dark, gloomy areas. For this reason, research on pedestrian reassurance suggests that a minimum illuminance and/or uniformity may be more reliable predictors of reassurance than average illuminance.¹³ The graphs in **Figure 3-5** show reassurance difference results plotted against average illuminance, minimum illuminance, and uniformity. The smaller the gap between daytime results and nighttime results, the higher the reassurance. See **Section 4.3** for additional lighting design recommendations that lead to pedestrian reassurance.

A limitation of the average-to-minimum uniformity ratio is that while it provides a measure of the quantitative

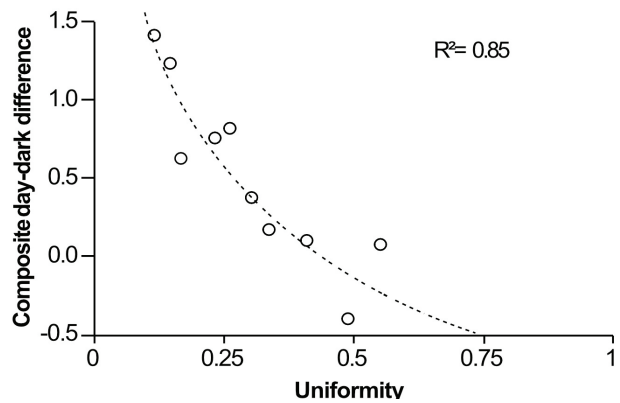
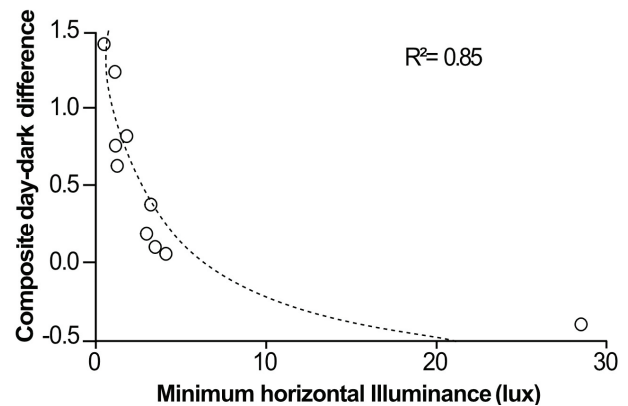
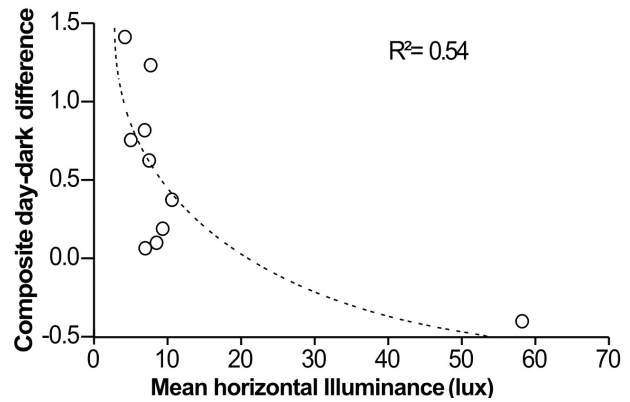


Figure 3-5. Minimum illuminance and uniformity exhibit a stronger connection to reassurance than average illuminance.¹³

change in illuminance, it does not characterize the rate of change. For the same uniformity, this could be a gradual or rapid transition from dim to bright areas, depending on the distance between successive luminaires and the optical control of the lighting system (see, for

example, **Figure 3-6**). If the rate of change is gradual, a pedestrian walking along a path has time to adapt to the change in ambient illumination, and the difference between areas of low and high illuminance may be less noticeable. If the rate of change is fast, the difference between areas of high and low illuminance will be more noticeable and may become uncomfortable.¹⁶ Increased awareness of this issue is especially important with the use of luminaires with more precisely focused light distribution, such as attainable with LED luminaires.



Figure 3-6. Uniformity alone does not regulate the rate of change from bright to dark. Shown here is a path with rapid graduations from bright to dark resulting in a spotty appearance. (Image courtesy of Rick Utting)

Note: European (BS EN 23201-2:2015) and CIE (CIE115:2010) standards use an inverse ratio to quantify uniformity, i.e., minimum-to-average illuminance. Therefore, the uniformity of a pathway with an average illuminance of 10 lux and a minimum illuminance of 2 lux is expressed as 5:1 in North America, and 0.2 in Europe.

3.5 Color of Light for Outdoor Pedestrian Spaces

Since the introduction, evolution, and acceptance of solid-state light sources, designers face new choices for specification. Chief among them are spectral power distribution (SPD), correlated color temperature (CCT), color rendering ability, and the broader general use of color.

When Howard Brandston contemplated how to illuminate the Statue of Liberty, his thought process led him to believe this lady of “green skin” looked best in the natural spectrum of dawn’s early light. The desire to mimic and repeat that scene during dark is based on the task of creating an emotional response from the viewer—hopefully, enjoyment—and warranted the effort and expense to spectrally tune electric light in an

era barely capable. The learned lesson is that spectrum and color might matter.

3.5.1 Spectral Power Distribution (SPD). SPD describes the variation across the visible spectrum of light output from a source. Light sources with different SPDs will exhibit variations in color appearance and color rendition. It is important that outdoor lighting for a public space support the types of tasks pedestrians wish to perform in that setting (see **Section 2.1 What We Need to See and Why**). The recommendations in this document assume adaptation luminance in a low photopic (5 to 10 cd/m²) or mesopic (1 to 5 cd/m²) range. In these domains, spectrum can influence peripheral detection of moving objects but does not have an effect on the detection of obstacles seen foveally.^{9,17-20} (Refer to *IES TM-12-12, Spectral Effects of Lighting on Visual Performance at Mesopic Levels* for further discussion.) However, emotion and enjoyment are also important goals for pedestrians in the outdoor environment (see **Section 4.1 Hierarchy and Context**). Seeing landscape, art, and other people realistic or otherwise may contribute to an emotional response goal. Therefore, SPD can be a key ingredient in exterior lighting design. Furthermore, emotion and feelings are also important in outdoor environments. (Refer to **Section 4.1 Hierarchy and Context** or **Section 4.5.2 Enjoyment** for design recommendations regarding the use of SPD.)

SPD also plays a significant role in the health of humans and the environment. The natural sleep pattern of humans, sky glow, the migration of birds, and the survival of baby sea turtles are all directly affected by the quantity and SPD of electric light. (Refer to *IES TM 18-18, Light and Human Health: An Overview of the Impact of Optical Radiation on Visual, Circadian, Neuroendocrine, and Neurobehavioral Responses* for the importance of SPD regarding human health,² and to *ANSI/IES LP-11-20, Lighting Practice: Environmental Considerations for Outdoor Lighting* for information regarding the effects of SPD on sky glow and the nighttime environment.⁴)

3.5.2 Correlated Color Temperature (CCT). White light sources, preferred for human vision and commonplace for lumen-based design, are typical for pedestrian spaces in the outdoor environment. The appearance of

white light can be characterized by CCT. This describes the appearance of near-white light on the absolute temperature scale, measured in kelvins (K). As shown in **Figure 3-7**, lighting of amber-white appearance is described as being warm white and has a CCT in the range of 2500 K to 3500 K. Neutral-white lighting ranges from about 3500 K to 4500 K and includes moonlight (4125 K). Lighting of blue-white appearance is described as being cool white and has a CCT of approximately 4500 K and higher. It is important to note that the differences seen in **Figure 3-7** are partially because the sources are lined up side by side. Seen individually, the apparent warmth or coolness diminishes with time as the observer adapts to the lighting.



Figure 3-7. Exterior pedestrian spaces commonly use white light with CCTs in the range of 2500 K to 4100 K.
(Image © SuriyaKK/iStock)

Adequate for appearance differentiation, CCT has become a commonplace and important specification within a project. However, CCT alone, often mistakenly used as a shortcut to explain spectrum, is an inadequate metric for many concerns important in the design of exterior public spaces, such as the resulting color rendering or impact on vision and human health. CCT is a unidimensional description of light, and it is possible for light sources with very different SPDs to have the same CCT. For this reason, it is important that the

lighting practitioner have an understanding of the SPD for a given light source to complement a CCT specification. (Refer to **Section 4.5.1 Atmosphere** for design recommendations regarding CCT.)

3.5.3 Color Rendering. The effect of SPD on the appearance of illuminated surfaces has, until recently, been characterized by the CIE general color rendering index (R_a or CRI). CRI has not historically played a key role in outdoor lighting design because light sources were chosen more for efficacy, cost, and maintenance regardless of their CRI values. However, newer long lasting and efficacious light sources with a broader spectrum make it possible to consider color rendering ability and how it might affect the pedestrian experience outside at night.

CRI describes the average difference in appearance of a small sample of standard colored surfaces when lit by a specific light source versus when lit by a reference light source. A CRI of 100 indicates that the colors are identical, while a CRI of less than 100 indicates differences in appearance but does not indicate which colors are different, the direction or magnitude of change, nor whether the change is likely to be perceived as better or worse. **Figure 3-8** shows examples of lighting with two sources, differing in both CCT and CRI. (Refer to **Annex C – Spectral Power Distribution and Color Rendering** for the limitations related to CRI and how ANSI/IES TM 30-18 can be used as a more comprehensive color rendering metric.)



Figure 3-8. The light sources in the image on the left have a CRI of 22, while those in the image on the right have a CRI of 81. (Image courtesy of Landscape Forms)

3.5.4 Use of Color. LEDs have put the use of color in lighting design within reach of many owners and have become a common discussion on projects that previously would have been hamstrung by cost, limited life span, obtrusive equipment, fragility, maintenance, and inefficient spectral filters. The challenge posed by the expansive possibilities of color is, simply, that just because one *can* use color does not mean one *should*. Educating clients on the use of 16.7 million possible colors is very important. What is the purpose of using color? Is the designer passionate about using color and can the client be convinced of its virtues? Does the client understand the advantages, and can they afford it? The answers to these questions are important because great effort goes into researching SPD, equipment, and mockups to meet the objectives.

If not applied carefully, color can work counter to the intended focus and become distasteful, crass, disturbing, or inappropriate, thereby diminishing the potential for success and doing more harm than good. Lighting designers will need to add the selection of light color and spectrum to their list of “standard care” and educate clients on the risks and rewards of using color. (Refer to **Section 4.1 Hierarchy and Context** and **Section 4.5 Atmosphere and Enjoyment** for design recommendations regarding the use of color.)

4.0 The Design Process for Pedestrian Lighting

Lighting is a complex, often subtle constituent of the nighttime environment. It can add value and meaning to objects and space, while supporting basic visibility for tasks. Quality lighting can help define a positive urban image and influence the decision of a pedestrian to visit, navigate, or engage with an outdoor space at night.²¹

The design challenges presented in lighting an outdoor pedestrian space can vary widely depending upon issues such as context, location, intended use, design, and even user expectation. Regardless of these often-varied characteristics, there are several fundamental

principles that contribute to successful illumination of outdoor pedestrian environments. Quality outdoor lighting should communicate visual order, hierarchy, and spatial character. Even specific placement of equipment can help determine much of the environment’s visual character after dark.

The design recommendations in the subsections that follow consider lighting for people in exterior environments throughout the public realm to ensure that the end result is planned, coherent, and satisfactory for the community. These recommendations may include lighting for architecture, cityscapes, sidewalks, pedestrian malls, pathways, bikeways, parks, monuments, buildings, statues, fountains, and landscapes. They also consider the larger environmental context of the community and transitions across multiple lighting zones. **Figure 4-1** graphically provides a hierarchical foundation and starting point, with a roadmap to designing quality lighting for people in outdoor environments.

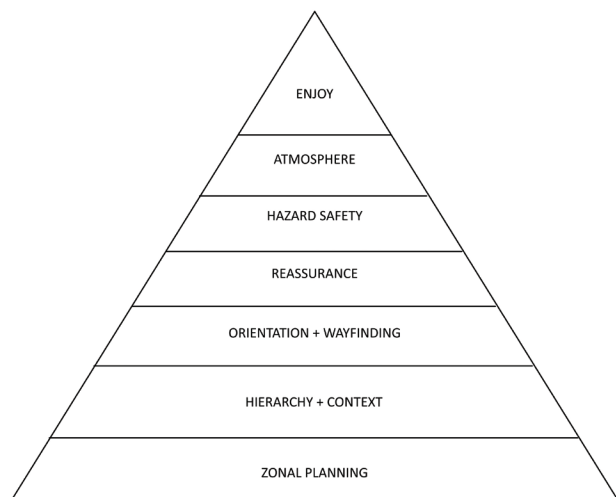


Figure 4-1. Hierarchical process of designing quality lighting for people in outdoor environments. (Figure courtesy of Landscape Forms)

Asking key questions to help identify the true project needs is one good way to begin the lighting design process. The following “thought starters” provide a beginning for an effective design process for exterior environments:

- What are the community goals? In what lighting zones are this project and the surrounding properties?

- What elements are most important and define the character of the space?
- What special points of interest (e.g., sculpture, landscape, amenities, exits) need to be lighted?
- What additional elements add to the general understanding of the space?
- How can CCT, SPD, and color rendering be used to accomplish the desired outcome?
- What are the visual task requirements? For example, are people walking, sitting, reading, socializing, relaxing, enjoying?
- How will luminance ratios affect pedestrian visibility? Can there be lighting patterns and strong contrasts, or should it be illuminated in a uniform way?

Although just a beginning, asking proper questions to more fully comprehend a project's challenges is fundamental to good lighting design. The following subsections provide guidance in using the answers to help establish a successful lighting design approach. Finally, and perhaps the most important tool prior to project implementation, will be physical mockups. Obtaining sample equipment and energizing it for critical review is the best way to evaluate the design, modify the plan, and convince the client or owner to make the investment.

4.1 Hierarchy and Context

A hierarchy of public lighting conveys the relative importance and character of cityscapes and enhances their information-giving value. Limitations and restrictions applied to public lighting can also impart information and create impressions about the character of the area. Lighting designers should be aware of the historical, cultural, and functional context of a building or space, as well as its architectural concept and structure. For parks and outdoor gathering spaces, these responsibilities extend to the lighting of landscape, vertical surfaces, paths, structures, and other key objects that, when observed and considered collectively, help establish a spatial framework and visual coherency. (See **Figure 4-2** for an example.)

Lighting designs should respect and relate to the principles, ideas, and intended hierarchical relationships of the visual environment. What elements are most

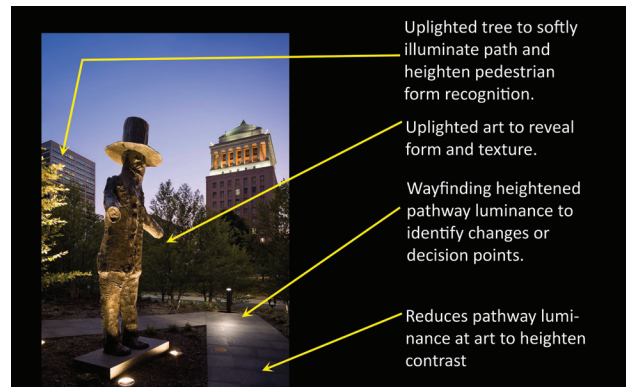


Figure 4-2. A spatial framework is created by coherent lighting of important and non-important elements. (© Debbie Franke)

relevant for establishing spatial comprehension and reinforcing the intended use and feel of the space? How should the composition of objects and their surfaces be illuminated in order to portray the intended (perhaps client mandated) hierarchical relationships? What is the contextual “dialog” between the space being designed and the adjacent environments? The answers should influence an approach to the design that includes appropriate balance and sensitivity. **Figure 4-3** provides an example.

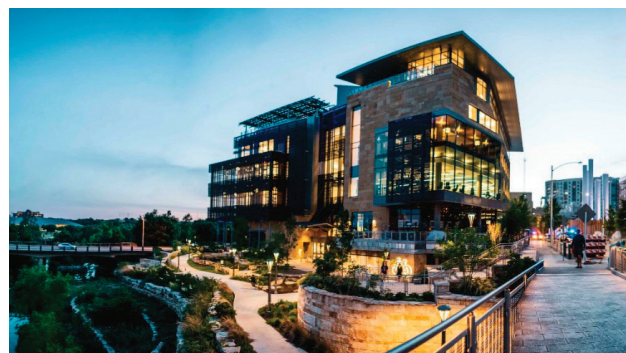


Figure 4-3. The exterior hierarchy highlights the landscaped entrance and natural façade materials, providing a visual connection with the adjacent parks and bike trails. (Image courtesy of Landscape Forms)

Street lighting has traditionally been the basic component of public nighttime lighting. In urban settings, it is the street lighting that, along with traffic signals and signage, organizes and defines the visual environment at night. The height, size, type, quantity, and location of equipment all contribute to

the lighting hierarchy. The streetscape and associated pedestrian spaces should appear consistent with the community focus and be appropriately integrated. The visual image is just as important during the night as during the day.

4.1.1 Defining Community Goals and Themes. The application of lighting zones can help establish the hierarchy of larger regions and define areas of activity and importance. Common themes in architecture and lighting can assist communities or special districts in establishing unique identities. Some communities may value the public view of the night sky, while others may be focused on a retail corridor—and the two need not be mutually exclusive. Good communication with owners and architects regarding community planning will help establish goals and methods for lighting. If possible, visits to benchmark installations are recommended to help establish a mutual understanding of the design objectives among designers, stakeholders, and decision makers.

For example, a rural mountain community may wish to limit the height, visibility, and brightness of lighting equipment to minimize light pollution and light trespass. This will help maintain a sense of quiet solitude in the wilderness, where few cars or pedestrians are expected. In contrast, the high volume of traffic and pedestrians in an urban entertainment district requires elevated light levels for both safety and security. To meet the higher lighting criteria, several different types of lighting equipment can be used: human-scale poles with low-glare optics providing downward lighting for pedestrian zones, taller poles for roadways and parking areas, and façade lighting on nearby architecture. With careful coordination of such issues as decorative detailing, light source color(s), and luminaire brightness, these separate elements can combine to provide a visual identity for the area.

Prioritizing community goals will help define the requirements of the outdoor lighting system. Some examples for consideration include:

- Identifying when not to light
- Showing sensitivity to environmental light pollution and light trespass, including reviewing adjacent areas for possible conflicts

- Determining the community's desired nighttime ambience and beauty through architecture and/or nature
- Considering crime prevention and pedestrian reassurance
- Understanding project economics, including energy usage and system maintenance

4.1.2 Transitions between Different Tasks, Zones, or Light Levels. Consensus recommendations provide guidance as to the quantity and quality of the illumination needed in a given area for a specific visual task. In reality, in any given area, people may be engaged in many different tasks. This is compounded by the fact everyone has unique visual needs. These can be age dependent, include a sensitivity to glare, or involve a physical or neurological condition. The designer needs account for these variations and use appropriate illumination to help guide people in their use of an outdoor space. As many lighting projects are site specific, it is always beneficial to evaluate adjacent spaces and their lighting systems so that transitions on- or off-site can be planned.

Lighting transitions are often needed between spaces with differing visual tasks and, typically, varied illumination levels. Lighting designers have two basic options for consideration. The first is an abrupt change in light levels, which would be most conspicuous, or even dramatic. These bright-to-dark transitions can make the sequence of spaces appear more interesting and may evoke emotions for the visitors or guests. The subtle use of multiple CCTs or even the bold use of color can create a dynamic transition.

More often, however, the intent is to create a subtle blend of light levels to make the transition zone visually disappear. This is made easier by the eye's inability to see subtle shifts in illuminance. Using **Figure 4-4** as an example only, this train platform required the project's highest level of illumination, 50 lux. Because 50 lux at night is so much higher than typical pedestrian requirements, transitional lighting was an appropriate technique to adapt pedestrians' visual systems to and from lower light levels. The first transition was in leaving the platform and entering the primary plaza, illuminated to 25 lux. A second transition then took place between



Figure 4-4. An example of a transitional lighting plan allowing pedestrians to navigate between different light levels. (Image courtesy of Nancy Clanton/Clanton & Associates)

the primary plaza and the secondary street or sidewalk, illuminated to 10 lux. On the other side of the project, the adjacent parking lot would typically need much less light, perhaps 2 to 5 lux. To facilitate adaptation and elevate the sense of reassurance, the transition zones needed to be gradual so that they visually melded into a single pedestrian zone, meeting the individual task needs of each space.

4.2 Orientation and Wayfinding

Orientation, the ability to understand one's place within one's surroundings, is one of the most fundamental the pedestrian needs after dark. Lighting practitioners can assist this process by revealing vertical objects and surfaces in a comprehensible manner. This *vertical and horizontal differentiation* can provide the pedestrian with a visual framework, making orientation easier—both consciously and subconsciously.²² Nighttime orientation is facilitated by looking to the horizon, to boundaries and vertical objects near and far, to both static and moving features, and to pathways. It is essential that lighting support this process of pedestrian observation and detection.

People orientate themselves during daylight with the aid of landmarks such as:

- Architecture, buildings, towers, bridges
- Sculptures, monuments, and similar hardscape

- Water features
- Landscape and flora elements

These objects help pedestrians create their “image of the city” or “sense of place.”^{23,24} Being deprived of landmarks during the night can make visual orientation more difficult. This document recommends appropriate uses of lighting at night to emphasize important visual features and their intrinsic characteristics, in order to aid pedestrian orientation.

As people gaze at near and far objects in the visual field, the relationships between illuminated vertical planes will influence their sense of orientation. The brightness of some surfaces may serve to reveal and model foreground elements through silhouette, where the visual task is darker than its background, to acknowledge significance or establish visual hierarchy. This is often more important than illuminating the pathway; vertical elements act as wayfinding markers and provide intuitive navigational guidance. Furthermore, the thoughtful management of luminance and brightness dissimilarities throughout the environment can assist in establishing a level of coherency to the surroundings.²⁵

Wayfinding, the ability to move and navigate in an environment, is both a conscious effort—especially in new or infrequently encountered surroundings—and a

subconscious one, as the eyes are constantly providing perceptual information to the brain. While boundaries may provide physical security from threats such as water hazards or vehicular traffic, they can also reduce pedestrian confidence if destinations or means of egress from a space are not obvious, near enough, or of sufficient quantity. Detection of these boundaries and the identification, or highlighting, of paths or means of egress will heighten pedestrian confidence, especially if both near and distant pathways are consistently revealed and identified.

A well-conceived, coherent lighting strategy becomes a solid cornerstone for pedestrian orientation and wayfinding after dark. It is important that the lighting of paths, floors, walls, buildings, and all manner of vertical surfaces are considered in the visual hierarchy of a design. Simplicity can create patterns and continuums that influence pedestrian understanding of a space. Complexity introduced by lighting that does not support orientation, wayfinding, or object recognition should be avoided.

All of these principles are visible in the left-hand photograph in **Figure 4-5**. There is a strong emphasis of the park border at the near and far horizon, illuminated in a consistent manner and thus clearly identifying the exit. In addition to the illumination of other vertical surfaces, there is a strong and simple repetitive pattern of pathway lighting, creating a contrast between path and the side landscape. Important art works are also illuminated, providing landmarks and a night image akin to the daylight version of the park.

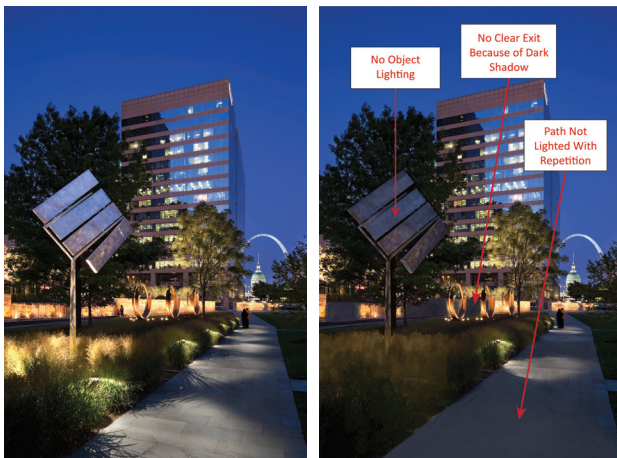


Figure 4-5. Pedestrian space with essential wayfinding principles (left) versus without (right). (© Debbie Franke)

In comparison, the park photo on the right in **Figure 4-5** is a graphically manipulated version showing how difficult orientation at night can be if lighting does not follow simple and strong principles. Lighting in this example does not emphasize the architectural and natural configuration of the space. If the park border at the rear is illuminated in an irregular way, visitors will have a more difficult time scanning for the exit. In addition, without repetitive pathway lighting or acknowledgement of the art works, there will be diminished understanding and wayfinding of the space. Visitors might anticipate special conditions where there are not any, and the park will not be appreciated for its uniqueness.

4.3 Reassurance

Reassurance, a subjective evaluation, is defined as that which makes someone feel less worried, less afraid, or less doubtful, and which restores confidence (see also **Section 2.1.2**). Much of the background research presented in this section uses the “day-dark” method, where data is captured during daytime and again after dark. Using this method, the effectiveness of electric lighting after dark is evaluated by calculating the differences between the two sets of results.¹¹ Quality lighting solutions will be those that minimize the day-dark difference.

For the purposes of this document, the reassurance task is an assessment of potential threat from other people, and the willingness of a lone pedestrian to engage with an outdoor environment after dark. In fact, reassurance can be influenced with lighting and has been found critical in a pedestrian’s willingness to engage with his or her surroundings.²¹ Ultimately, it is this engagement, along with other pedestrians, that activate an outdoor space.

It is important to note that if a space is unsafe during the day, lighting conditions at night are unlikely to make it safer. It is expected that an outdoor space at night will be less reassuring than in the daytime—what quality lighting can do is minimize the reduction in reassurance after dark. Quality lighting is often ranked as essential to pedestrians’ belief they are better equipped to cope with, and solve, visual tasks at night.²⁶ Research of reassurance using the day-dark approach has shown

that quality lighting at night makes it possible for an area to be only slightly less reassuring than during daytime.¹³ Thus, quality lighting can reduce the degree to which reassurance deteriorates after dark.

Vital for pedestrian reassurance at night is the ability to detect the presence of other people and activity in the periphery. When boundary and peripheral illumination are absent and the spatial perimeter is visually uncertain, the feeling of well-being and perception of safety may be compromised. Vertical illuminance becomes extremely important for this task. Illuminated vertical planes formed by plant materials, objects and surfaces along pathways help define and soften surroundings, and typically aid in depth perception and increased feelings of comfort and well-being. **Figure 4-6** provides an example.



Figure 4-6. The image on the left has a sharp-edged horizontal cutoff. The image on the right includes a softer edge and vertical illumination to provide greater understanding of the surroundings. (Image courtesy of Landscape Forms)

It is essential to provide an opportunity to become informed of possible threats, such as other people, as early as possible so that there is an appropriate amount of time to make decisions and react. This requires not only a line-of-sight view of the possible threat, but also the identification and status of surrounding boundaries and egress points. As discussed in **Section 4.2 Orientation and Wayfinding**, pedestrian confidence is not abetted if surroundings are misunderstood and exits from a space are not obvious, near enough, or sufficient in quantity. Lighting a wall or other vertical feature may increase the opportunity for the recognition of approaching pedestrians at distance; the comparative images in **Figure 4-7** provide an example. Threat

assessment decisions can be lifesaving when exits are easily identified, or when shelter and aid opportunities are clear and apparent. It is therefore not “wasteful” when exit and special refuge spaces are emphasized with additional lighting.



Figure 4-7. A pedestrian space that supports pedestrian reassurance (left) versus one without (right). (© Debbie Franke)

As stated, “other people” can be a primary threat to pedestrian reassurance. A diminished ability to detect other people, or threatening activity, can greatly diminish the pedestrian experience—often to the point of avoiding a public space altogether. At what distance can someone determine whether another person or group of people represents a possible threat? What, if anything, are they holding? Can their emotional state or intent be determined? When encountering other people in public space, it is essential to have the ability to assess threatening appearances or behaviors early, in order to make the decision to approach or avoid. Research suggests that a combination of facial expression recognition, gender and body type, hands and posture, and group size will produce the most informative results, with facial expression recognition (i.e., recognition of emotion as conveyed by facial expression) being the most informative.²⁶

4.3.1 Facial Expression Recognition. A primary parameter to create a feeling of reassurance is to illuminate the faces of approaching strangers, or strangers in the mid to far field of view. Human faces are key to a pedestrians’ appraisal of other people. A face can convey information about age, gender,

identity, emotional state, and possible behavior or intent. Research uses facial expression recognition as a standardized task to measure the effectiveness of lighting in conveying this information.^{10,26} Vertical illumination matters for facial evaluation. When facial features such as eyes, mouth and eyebrows are obscured, then facial appraisal becomes more challenging. While line-of-sight illuminance will vary, light—direct or reflected—should come from all sides and be relatively continuous, in order to ensure the possibility of seeing an approaching person's face.

The time available for useful and reliable facial appraisal is very small. Gaze duration and observation distance for this task play a key role in the illumination recommendation. One tacit basis of current guidance is that the minimum distance for making an evasive decision is 3.7 meters (12 feet). However, analysis of pedestrians' gaze behavior using mobile eye tracking suggests a tendency to look towards other people for only very brief glances (typically 0.5 seconds) and when they are at a distance of about 10 to 15 meters (33 to 49 feet).²⁷ Therefore, vertical illuminance should facilitate interpersonal facial recognition at that distance. Facial assessment at this distance would be a positive for giving the observer more decision and avoidance time. Unfortunately, the increased gaze distance affects the visual size of the target, and thus the ability to detect facial detail is more challenging.²⁸ For this reason, expense and over-lighting are serious concerns towards making facial details recognizable at distance. In fact, the additional illumination could actually present itself as glare and become counter-productive, preventing facial recognition. To prevent this condition, background illumination should not be more than four times higher than the target.

4.3.2 Other Visual Cues. Real-life public situations are complex and dynamic. A walking person's face and body change position with respect to the surroundings, and movement changes the position of other people relative to light sources. Thus, the scene of the observer changes.

Although less discriminating, gender, body type, posture, and hand gestures also provide information regarding threat assessment; their requirements for

lighting are less demanding than those for facial recognition. Expectation and familiarity can help in evaluation of the gender and identity of people from movements such as speed, rhythm and amount of arm swing, or gait and stride length. In these cases, the strategy can involve less surround lighting because vertical illumination in the immediate horizon will do an excellent job presenting information via silhouette. If this is the strategy, silhouette lighting needs to be continuous, i.e., without interruptions or dark space (see **Figure 4-8**).



Figure 4-8. Illuminated vertical surfaces can provide silhouette, increasing visual cues and aiding in threat assessment. (Image courtesy of Rick Utting)

4.3.3 Security vs. Reassurance. Security is never guaranteed, but neither is it acceptable to label every space as a security problem. Security problems are not predicated on lighting; they also happen in highly illuminated LZ-4 environments (see **Section 2.4.1** for a description of LZ-4) and in daylight. However, it is prudent for risk management and crime prevention experts to interpret space conditions on behalf of those with primary control and responsibility. Security lighting is intended to contribute to the protection of people and property from criminal activities. For applications where security is determined to be a major issue, due diligence activities and lighting guidelines may be found in *IES G-1-16, Security Lighting for People, Property, and Critical Infrastructure*.²⁸

4.4 Pedestrian Safety

Once lighting has provided for pedestrian orientation and a sense of reassurance, engagement with a public space should be anticipated and safe navigation should be the goal. This is an important step in the hierarchy of outdoor pedestrian lighting design because the cost of

injuries from terrain-related trip and fall accidents are a major economic concern for North America and abroad. Such hazards are responsible for most non-vehicular pedestrian accidents.^{29,30}

Records show that pedestrian trip and fall accidents are more frequent among the elderly. The elderly are more vulnerable due to their reduced peripheral vision and slower balance recovery times.¹⁷ Nighttime navigation of outdoor terrain does not make their task easier. Approximately 40% of Americans over the age of 65 fall at least once a year, and half of these accidents occur outdoors.⁹

4.4.1 Terrain Hazards. *Terrain hazards*, for the purposes of this document, are potential hazards within the pathway ahead caused by changes in surface or grade. Stairways, curbs, wheel stops, raised pavement, potholes, and slippery surfaces are all examples of possible trip and fall hazards in the outdoor environment. It is critical that pedestrian safety include the ability to detect and avoid potential terrain hazards. Knowing their limitations, the elderly may actually select a less disturbed and more certain path rather than the shortest one in order to avoid difficult obstacles.³¹

When finding their way in the outdoor environment, pedestrians scan the terrain in front of them to identify possible paths of travel. In some cases, there may only be one option. Based on eye tracking research, Fotios and Uttley suggested that pedestrians tend to scan for these trip hazards 3 to 4 meters ahead, which provides enough distance to change gait if an obstacle or terrain disturbance is detected.¹⁷ While it was previously believed that a 25-mm (1-inch) obstruction was the critical height for a trip hazard, in analyzing accident and foot clearance studies, the researchers concluded the critical height should actually be 10 mm (0.4 in.), providing a much more difficult job for the lighting. Using the new criteria, previous lab studies,^{17,18,20} and an emergency-lighting study,³² it was concluded that in the absence of pedestrian glare, 1 lux is sufficient illumination for pedestrians of all ages to safely detect trip hazards under any light source.¹⁷

Lighting after dark should create enough *identification* and *differentiation* between multiple pathway choices

to give pedestrians an understanding of likely trip and fall risks before navigation. Stairs, curbs, and pathway splits are necessary and expected hazards that contribute to our decisions about navigation. Purposeful lighting and insightful use of surface materials can help reveal these hazards (see **Figure 4-9**). However, in addition to providing enough information for the selective navigation of these known hazards, lighting should illuminate and reveal the pathway terrain itself, allowing pedestrians to avoid unexpected hazards such as manhole covers, poorly maintained pavement, slippery surfaces and other physical obstacles.



Figure 4-9. Quality lighting allows for navigation decisions when presented with multiple pathways. (Image courtesy of Landscape Forms)

Research shows that detection of moving objects within the peripheral vision increases and plateaus in the absence of glare and at an illuminance of 2 lux.⁹ The designer needs to decide where the light source(s) will be positioned and select equipment with a suitable glare rating and distribution type.

Finally, with an understanding the design objectives and adjacent lighting conditions, the designer should determine an appropriate rate of transition so that light sources can be spaced properly to meet the design objectives.

There are two special conditions that should be considered here:

- Lighting for stairs or other changes in elevation should be designed to minimize glare for the pedestrian and preserve tread and riser visibility,

such as may be attained with consistent shadows (see **Figure 4-10**). Creating complete visual uniformity on treads and risers made of like colors and reflectance may not provide enough contrast for safe navigation.

- Curbs and wheel stops can be detected as possible pedestrian trip and fall hazards under low light levels if the pavement and obstruction reflectance are in contrast with each other and if the illumination source contains a broad enough spectrum. Research by Bhagavathula and Gibbons concluded that visual performance for these tasks plateaued at 2 lux for walkway and curb visibility in parking lots.³³

4.4.2 Object Hazards. *Object hazards*, for the purposes of this document, are static or moving objects not normally characteristic of the terrain. Outdoor furniture, fallen tree limbs, and pedestrian debris are all examples of static object hazards. Cyclists and skateboarders, who share many of the pedestrian pathways, are examples of potential moving-object hazards. Pedestrian safety includes the ability to identify and avoid object hazards.

As explained in **Section 4.4.1 Terrain Hazards**, research shows that pedestrians fixate and gaze at their pathway for a significant portion of their travel time.³⁵ Meanwhile, the remainder of the visual scene is mapped using peripheral vision. For this reason, it is important to note that unexpected and moving objects may first appear within peripheral vision.

In addition, the overall pedestrian environment should be considered when defining light levels. The task area of the pathway should not be over-illuminated, as this might preclude pedestrians' experience of the ambient feel of the environment. Though the human visual system adapts relatively quickly to most light levels, contrast is an important factor in the identification of an object. *Note:* When introducing a significant light level difference into the field of view, the designer should be aware of the visual adaptation that will occur and the consequences for the visibility of other objects. Sometimes lower contrast ratios are prudent to permit a better overall visibility condition.

Recommended object safety illuminances are provided in Table A-1 in Annex A of ANSI/IES RP-43-20.¹⁵

4.5 Atmosphere and Enjoyment

As previously discussed, pedestrians have a hierarchy of visual tasks that nighttime lighting needs to facilitate. Once lighting has provided for pedestrian orientation, reassurance, and safe navigation, then and only then, should the lighting designer work to create the important but less tangible feelings of "atmosphere" and enjoyment.

The immediate luminous environment has an effect not only on task performance and reassurance, but on the emotional responses to and perception of a space. Oftentimes, lighting designers are required to balance these different goals. For example, higher uniformity across a task surface may enable *efficient* execution of a task; however, a lower uniformity with interesting contrasts might be perceived as more comfortable, and thus provide a more attractive luminous environment



Figure 4-10. Changes to terrain, such as stairs, should present enough contrast for safe navigation. (Images courtesy of Nancy Clanton/Clanton & Associates)

while simultaneously providing appropriate lighting for *reasonable* execution of the task.

The subsections that follow provide guidance intended to encourage discussion about how different design approaches may support appropriate task lighting solutions while also creating atmosphere and enjoyment.

4.5.1 Atmosphere. Several factors contribute to the “atmosphere” of an exterior lighting design. These include glare, uniformity, light levels, light source color, and the lighting equipment.

Glare reduction. Lighting design may vary greatly, even given identical objectives. One main differentiator is the amount of directional light used, or encountered from off-site sources, versus diffused or reflected light. The presence of glare in an outdoor pedestrian space erodes the visual interpretation of the surrounding atmosphere. It is essential to take measures to reduce both discomfort and disabling glare to the greatest extent possible. This is perhaps the most important duty of lighting practitioners in the design of exterior public spaces. Understanding and maximizing access to important spatial cues throughout the environment is best achieved by the elimination of glare.

Glare reduction and mitigation can be accomplished through lower light levels, dimming, luminaire optical strategies such as diffusion or redirection, regressed or shielded sources, luminaire location and mounting height, and careful understanding of the luminous intensity at all angles within the field of view. **Figures 4-11** and **4-12** show examples. Further glare

reduction can include the recommendation of outdoor environments using less specular materials, finishes like natural wood and stone, and landscaping.

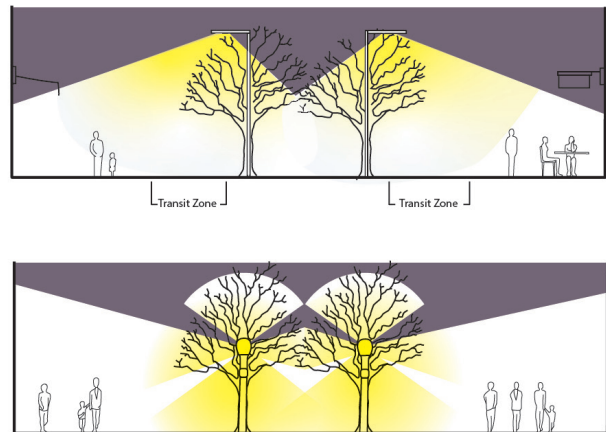


Figure 4-12. Mounting heights and luminaire position can reduce glare by changing the angles of emitted light within the pedestrian field of view. (Images redrawn from originals by Clanton & Associates, Inc.)

Uniformity. The uniformity of illumination is customarily and properly linked to visual performance and related issues such as object identification and terrain safety. The role of uniformity in creating an appealing pedestrian pathway is perhaps less obvious. In practice, soft-edged gradients may be preferable to sharp-edged pools of light. While higher light levels and uniformity are critical in some situations, such as stairs and elevation changes, they are less critical for wayfinding. The lighting designer may be creative and employ layers of light to illuminate vertical objects, edges, and the complete surround of perimeter illumination (see **Figure 4-6** in **Section 4.3 Reassurance**).



Figure 4-11. Lower light levels, luminaire diffusion, and lower mounting heights all reduce glare. (Images courtesy of Landscape Forms)

Light levels. Well-designed pedestrian spaces avoid over-illuminated task surfaces while still meeting code mandated values. Lighting designers should target using just enough light for the task and no more. This saves energy while also supporting a nighttime adaptation level—important for meeting pedestrian visual needs. An enjoyable walkway should have light levels related to the spatial context, as well as uniformity appropriate to the visual rhythm of the adjacent softscape, hardscape, and architectural features. For this reason, light levels should not be constrained into an absolute value of illuminance. In providing comfortable, well-orchestrated variability in light levels across a site the designer should consider adjacencies and zonal precedents to help foster intuitive orientation and wayfinding cues (see **Figure 4-13**).

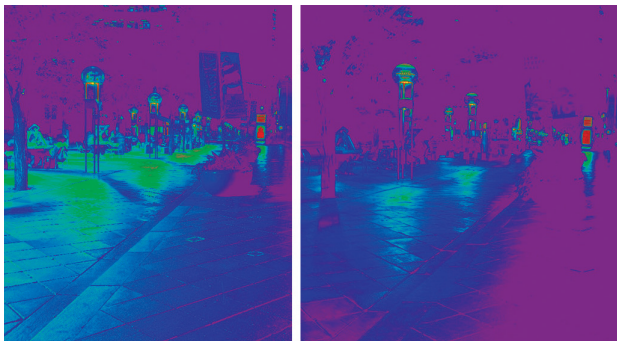


Figure 4-13. The HDR image on the left shows higher light levels and a broader range of uniformity compared to the image on the right. (Images courtesy of Clanton and Associates)

Light source color. Light color choice can make a significant impact on spatial understanding and aesthetics. There are often two design drivers in the selection of CCT. First is the cultural and contextual background of the region, community, and neighborhood. Is there an instinctive or indigenous influence where a culture or region prefers warm-white light versus cool white? Studies and surveys support the concept that CCT preference is not universal and may be partially based on geographic region, city, or even gender.³⁴ Clanton et al. conducted field studies in four U.S. cities with responses from more than 400 people. In two cities, differences in CCT led to different preferences (in this case, a preference for 3500 K compared with 5000 K), but in the other two cities no preference was found.

A second design driver is the context of the architectural style, texture, materiality, reflectance, and function or

use. Light source color(s) and color rendering properties influence how an environment is perceived. It is very typical for lighting designers to use warmer-white colors on historic landmarks to emulate the look and feel of historic gas or incandescent lamps associated with the period. It is also common for warmer-white colors to be used when humans are involved to evoke feelings of comfort and render scenes as less institutional. Cooler-white color(s) may be used to differentiate a project, accent nearby elements, or provide visual hierarchy cues.

Equipment. Another important consideration in creating the desired atmosphere is the height and position of the lighting equipment. The height of luminaires is important within the pedestrian space during both day and night. Luminaire heights that position the light source nearer people's heads instead of high above them help to create a more personal, human-scale lighting system. In general, lower positioned light sources help to produce and reinforce a more human-centric, intimate, and calming atmosphere. When sources are taller than 5 meters (16 feet), the resulting luminous environment can feel vehicular-centric and utilitarian, compared to an environment with light sources or luminaires that are lower (see **Figure 4-12**).

4.5.2 Enjoyment. Pedestrian enjoyment outdoors after dark is the primary goal of the guidance provided in this Lighting Practice document, and arguably may be the biggest compliment for a lighting designer. As discussed previously and shown graphically in **Figure 4-1 (Section 4.0)**, lighting design considers a hierarchy of visual tasks and objectives that need to be met before enjoyment can occur. In other words, enjoyment cannot happen if there is not a positive atmosphere where pedestrians feel reassured of their surroundings, and reassurance cannot happen unless there is understanding of how to orient oneself within and navigate an outdoor space without physical risks. Further, navigation and object detection cannot happen without proper illumination based on the ambient lighting conditions of adjacent property and zonal planning. It is assumed all these hierarchical prerequisites will be met before advanced lighting design techniques targeted at creating enjoyment are attempted.

Originality. Discerning lighting design can contribute significantly to aesthetic appreciation of culture and

diversity. However, new and extraordinary lighting designs can become a contribution to culture—a powerful argument for the adoption of a very special lighting design. A successful design will provide sufficient and necessary contrast for spatial orientation and directional navigation without the introduction of glare that could hinder visual comprehension.

Pedestrians are stimulated by contrast, non-uniformities, and the “play of brilliants” (addressed further down in this section) in the visual environment. Examples of such non-uniform (but nonetheless well-liked by the public) environments may be found in many downtown nightlife districts, public event plazas, urban public parks, and college campuses.

Elements of enjoyment. Some pedestrian tasks naturally associated with enjoyment. Play, entertainment, viewing nature, and appreciating art are all occasional tasks capable of bringing spontaneous pleasure. Illumination of architecture, sculptures, monuments, water features, and landscapes are not only important opportunities for meeting basic pedestrian needs like orientation and understanding; they are also great opportunities to create atmosphere and enjoyment (see, for example, **Figure 4-14**). Specific illumination recommendations for these kinds of tasks can be found in the narrative and tables of *ANSI/IES RP-43-20, Recommended Practice: Lighting Exterior Applications*.¹⁵

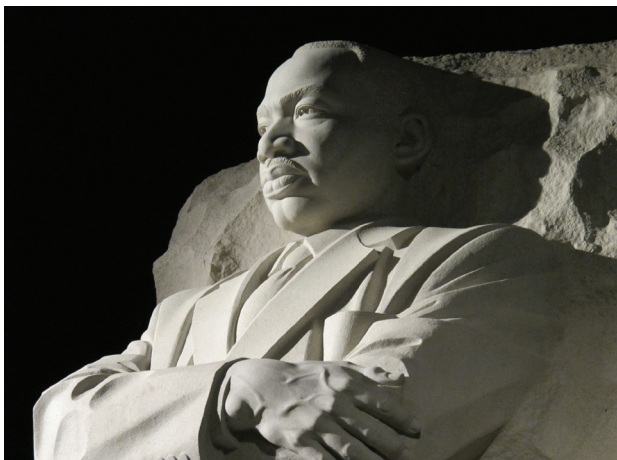


Figure 4-14. Careful manipulation of light intensity, direction, and distribution can enhance surface features that reinforce interpretation of a visual scene. (Image courtesy of Randy Burkett Lighting Design)

SPD enhancement. The visible region of the electromagnetic spectrum extends from 380 nm to 770 nm. Different types of light sources emit radiation in different amounts across the wavelengths of the visible spectrum. Within this range there are multiple ways for the various wavelengths to combine and make white light. Ultimately, the many variations in spectral power distribution (SPD) are another tool allowing lighting designers to have an impact on the visual perception of outdoor spaces. Pedestrian outdoor malls, landscapes, and retail spaces can be enriched through careful consideration of SPD options. Lighting a green landscape space with a green-deficient spectrum might make the trees and foliage look sick, while illuminating it with a green-rich spectrum can make it look vibrant. SPD enrichment is not limited to a mono-chromatic boost. In a cityscape with combinations of hardscape, furniture, plantings, and people, the variations in materials, color, texture, and reflectance are vast. Using a rich, broad-spectrum light source may be the best way to ensure that people and plants look healthy in a scene properly rendered (see **Figure 4-15** for an example).

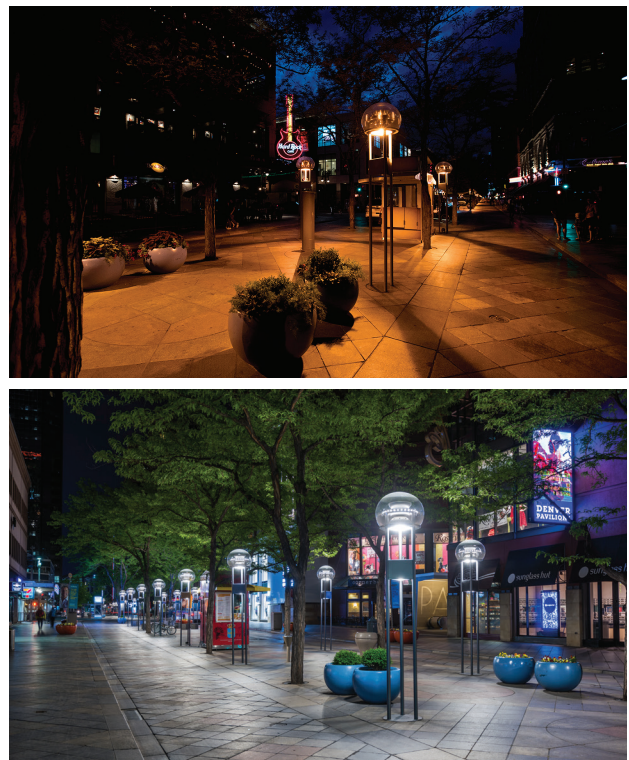


Figure 4-15. “Before” image on the top. “After” image on the bottom, showing glare control, low light levels, effective uniformity, and excellent color rendering to improve the visual experience. (Image courtesy of Landscape Forms)

Color (hue and saturation). Color is not typically the main source of pedestrian lighting, but rather a complement used for accent, identity, branding, and mood creation. The amount of added color used should reflect the idea of an accent, third in priority (behind ambient and task lighting) so it does not become overwhelming and work counter to its intent. **Figure 4-16** provides a good example.



Figure 4-16. The use of colored light in reinforcing architectural features can amplify distinctiveness and reinforce intended atmosphere. (Photograph courtesy of Landscape Forms)

The use of color is often assumed to be dynamic and changing; however, it is frequently static. Ideally, the use of color is deliberate and obvious, but sometimes its influence is happenstance. Color can call attention, provide information, and celebrate a season, holiday or special event. Warm-toned, longer wavelength light is often associated with relaxation, comfort, coziness, and feelings of privacy. Cool-toned, shorter wavelength colors can enhance a sense of public space, openness, crispness, and institutionalism in appearance.

The inclusion of dynamic lighting can bring excitement and animation to enhance the visual experience; however, it requires a thoughtful approach. As humans react toward movement and change in the visual field with priority, it is essential to consider dynamic lighting wisely and not create an unnecessary, distracting, or even misleading visual message that could compromise spatial relationships. A slowly changing lighting effect, whether by intensity or color, can often be more appropriate and acceptable than a rapidly fluctuating one. The impression of such dynamic effects should be

considered in both their near- and far-field contexts in order to completely understand their overall effect on the environment.

The way that dynamic lighting effects are perceived by a casual observer might be different from how someone with a continuous view will perceive them. For example, the experience of walking past a dynamic display may be pleasurable, but it might be irritating for someone living across the street. Dynamic lighting can also be a distraction for motorists, which is hazardous. All potential observers need to be considered in the design. Existing and newly created visual hierarchies should be well understood and respected. As there is also an effect of adaptation, lighting designers should ask themselves how a dynamic concept can be designed to remain both relevant and impactful over time. Although current and emerging lighting control technologies make it relatively straightforward to create dynamic lighting effects, the true impact of those effects on an outdoor space need to be well understood when they are applied.

When implementing a new pedestrian lighting scheme involving color, it is essential that designers work with mockups and demonstrations to achieve the preferred outcome of most stakeholders. Engaging in a public dialog will provide people with the right language to express their preferences. If resources are scarce to complete such work, designers can refer to the studies cited in the **References** section and confidently provide their recommendations based on the cultural context and lighting zone for which they are developing a specification.

Play of brilliants. “The play of brilliants excites the optic nerve, which in turn stimulates the body and spirit and charms the senses. It creates feelings of aliveness, alerts the mind and awakens curiosity, and sharpens the wit. Play of brilliants quickens the appetite and heightens all sensation. It can be distracting, or it can be entertaining.” – Richard Kelly

When programmatically and aesthetically appropriate, nighttime pedestrian spaces may benefit from the introduction of small exposed light sources (emitting less than 75 total lumens) capable of providing *sparkle*

or *brilliance* through both their individual presence and collective assemblage. Creative application of such exposed sources can foster a sense of excitement and festivity as they draw attention and add visual emphasis to gathering areas or programmatic focal points. These effects are an often unplanned and instead serendipitous happenstance, such as the attractive and exuberant lighting from a nearby storefront, restaurant, holiday display, or self-luminous artwork (see **Figure 4-17**). However, great care should be taken to ensure that “sparkle” light sources do not significantly contribute to uplight and are not perceived as glare.



Figure 4-17. A random arrangement of grade-recessed LED sources sparkle in the visual field, providing a compelling compliment to the nearby water cascade. (© Patrick Schneider Photo)

5.0 Recommendations for Pedestrian Lighting

Specific illumination recommendations for pedestrian visual tasks in exterior environments can be found in the narrative and tables of *ANSI/IES RP-43-20, Recommended Practice: Lighting Exterior Applications*.¹⁵

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ADDITIONAL READING

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Annex A – Glare

Disability glare:

Glare is not just a problem because it is uncomfortable, or because it shifts the observer's adaptation such that details in the shadows are lost. Glare is also a problem because it reduces the contrast of objects in the central field of view. Bright light sources close to the axis of view actually scatter light inside the ocular media. The lens, aqueous humor, and other elements in the eye are not completely clear optical media, and any diffusion scatters light across all areas of the retina. It is similar to passing a narrow flashlight beam through a piece of gauze. The resulting beam exhibits an edge that is less sharp, more light reaches the area beyond the edge of the light beam, and the intensity of the central light beam is reduced. Scattered light in the eye is called *veiling luminance* because it is like laying a transparent veil over a paragraph of print. The print appears to be smeared as the contrast of the print is reduced. This reduction in contrast is called *disability glare*, and it is an problem related to task visibility rather than visual comfort, although a bright light close to the field of view may produce both at the same time.

Discomfort glare:

Glare can be uncomfortable, distracting, or even painful, even if it does not reduce visibility or affect adaptation luminance ranges. Multiple attempts have been made to quantify discomfort glare for outdoor applications, but one promising attempt uses the Unified Glare Rating (UGR), which has been used to predict discomfort glare for interiors for about 20 years. UGR has many limitations but is accepted in the International Commission on Illumination (CIE) as the best of the options.

The reason that the UGR formula falters at high luminaire position angles found in pedestrian lighting (i.e., above the visual field) is due to the Guth Position Index, the value of which grows very large above 55°, and results in the UGR value dropping very low. If UGR is to be modified to accommodate viewing angles at and above 55°, one likely approach is to modify the calculation for the position index.

The human visual field extends to about 55° above the point of fixation (assumed to be parallel to the ground plane), 70° below, and slightly more than 100° both to the left and the right of the point of fixation. This varies slightly according to an individual's facial structure. If a light source is producing discomfort glare, it was until recently assumed that the source was within the visual field and that its luminance was imaged on the retina. Once the light source was positioned outside that field, it theoretically no longer caused discomfort.

Experience by researchers and the design community suggests otherwise. When the sun is nearly overhead, wearing a hat with a brim clearly improves comfort (and clarity of the visual image, although that is disability glare). Similarly, shielding one's eyes from a bright electric light source overhead improves both comfort and clarity of the visual image. Furthermore, for most applications the observer's fixation point has been assumed to be parallel to the ground plane, but there are many situations where that is not the case, such as glancing around outdoors for street signs; or walking a path or neighborhood at night illuminated by pole-mounted luminaires. Sources of glare traditionally assumed to be out of the field of view may be spatially shifted closer to the axis of view. Consequently, research is needed to document whether light sources located outside the classical field of view need to be considered in the UGR calculation.

Another shortcoming of using UGR for outdoor lighting is that the basic equation assumes that the luminaire is uniform in luminance across its aperture. With the introduction of LEDs, many (if not most) outdoor luminaires utilize arrays of LEDs, and each individual dot of light seems to be perceived as more glaring because its luminance is often 500,000 cd/m² or higher. Joint Technical Committee 7 (JTC7) of the CIE has proposed a modification to UGR for use with luminaires

with non-uniform luminance distribution. In effect, the proposed modification suggests a high-dynamic range camera image be used to evaluate the luminous area exceeding 500 cd/m² from the luminaire from one or two typical viewing angles. That area is adjusted with a point-spread function, the areas are summed, and that becomes the luminaire's effective luminous area. Research is needed to show that this modification to the equation is predictive of glare from luminaires used in outdoor pedestrian-dominated spaces, but the concept is encouraging.

Additional information on glare may be found in *ANSI/IES LS-8-20, Lighting Science: Vision – Perceptions and Performance*.³⁵

Annex B – Spectral Power Distribution (SPD) and Color Rendering

The IES has recently issued *ANSI/IES TM-30-18, IES Method for Evaluating Light Source Color Rendition*.³⁶ This Technical Memorandum documents a comprehensive method for evaluating light source color rendition. It is a revision of IES TM-30-15 and contains three modifications that achieve consistency with the recent publication of CIE 224:2017, from the International Commission on Illumination (CIE).³⁷ CIE 224:2017 defines the General Color Fidelity Index (R_f), which is based on the Fidelity Index (R_f) calculation within IES TM-30-15, but with a few minor changes. These changes involve the extrapolation method used for the color evaluation samples (CES), the transition zone for the reference illuminant, and the scaling factor for fidelity calculations. These modifications cause only minor differences in the rank order of R_f values of light sources, and only have a small effect on the values for typical light sources. With the issuance of ANSI/IES TM-30-18, the two measures have been harmonized so that the R_f value in each is identical to that of the other, which is a key step toward broader acceptance of the metric. While this color fidelity measure gains acceptance, the designer may refer to more traditional color rendering metrics, but should also use the superior metrics established in ANSI/IES TM-30-18.

Annex C – Spectral Power Distribution (SPD) and Object Detection

Three experiments were conducted to examine the detection of pavement hazards in peripheral vision, using a range of observer ages, hazard locations, and experimental designs.^{9,18,20} These experiments show that there is negligible benefit to increasing light levels

above 2 lux and that age and SPD have no effect. Age and SPD become important at illuminances below approximately 1 lux. At such levels, older observers have poorer detection than younger observers; below 1 lux, detection is improved by using light source with higher S/P values. Interpretation of these data for the detection of a critical hazard (a 10-mm change in vertical height, as indicated by foot clearance data and injury records) suggests the optimal light level to be 1 lux regardless of spectrum.¹⁷

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Process for Change to an ANSI/IES Standard under Continuous Maintenance

This standard is maintained under continuous maintenance procedures, for which IES has an established and documented program for regular publication of addenda or revisions, including procedures for timely, documented, consensus action on requests for change to any part of the standard. Committee consideration will be given to proposed changes by June 30 of any given year for proposed changes received by the IES Director of Standards no later than December 31 of the previous year.

Submittal Format

Proposed changes must be submitted to the IES Director of Standards in the announced published format. However, changes may be accepted in an earlier published format, if the differences are immaterial to the proposed change submittal. If the Director of Standards concludes that a current form must be utilized, the proposer may be given up to 20 additional days to resubmit the proposed changes in the current format.

Specific changes in the text or values are required and must be substantiated. Any change proposals that do not meet these requirements will be returned to the proposer. Supplemental background documents to support changes submitted may be included.

Submission to the Committee Chair

The Director of Standards shall forward proposed changes received on appropriate forms to the committee chair for assigning to committee members (responders) to develop responses to submitters of proposed changes.

Review and Clarification

Responders shall review proposals and should contact the proposer if necessary for clarification.

Response Recommendation

Designated responders shall draft a recommended committee response, including any recommended changes to the standard. The 'responders' recommended responses shall be submitted to the committee chair in electronic form usable by Society Staff, including any recommended change to the standard in response to proposals received.

Options for Committee response are limited to:

- a) Proposed change accepted for public review without modification
- b) Proposed change accepted for public review with modification
- c) Proposed change accepted for further study
- d) Proposed change rejected

The responders shall provide reasons for any recommendation other than option (a) above.

The designated responders shall not recommend option (c) unless the further study can be completed by October 1 of that year, and providing the Committee can then vote for option (a), (b), or (d) no later than November 15 of that year.

Editing

The Committee chair or his or her designee shall edit the draft responses and circulate the edited drafts to the committee for review.

Form for Proposing Change to an ANSI/IES Standard under Continuous Maintenance

NOTE: Use a separate form for each comment. Submit to the Director of Standards, IES, 120 Wall Street, 17th Floor, New York, NY 10005-4001. Email: standards@ies.org. Fax: 212-248-5017.

1. Submitter: _____
Affiliation: _____
Address: _____
City: _____ State: _____ Zip: _____ Country: _____
Telephone: _____
Fax: _____
E-mail: _____

I hereby grant the Illuminating Engineering Society (IES) the non-exclusive royalty rights, including non-exclusive rights in copyright, in my proposals. I understand that I acquire no rights in publication of the standard in which my proposals in this, or other analogous, form are used. I hereby attest that I have the authority and am empowered to grant this copyright release.

Submitter's signature: _____ Date: _____

2. Title of publications and year published _____

3. Clause (section), sub-clause or paragraph number; and page number: _____

4. My proposal (check one):

- ☐ Change to read as follows
- ☐ Delete and substitute as follows
- ☐ Add new text as follows
- ☐ Delete without substitution

Use underscore to show material to be added (added) and strikethrough for material to be deleted (~~deleted~~). Use additional pages if needed.

5. Proposed change:

6. Reason and substantiation:

Select as applicable:

- ☐ Additional pages are attached. Number of additional pages: _____
- ☐ Attachments or referenced materials cited in this proposal accompany this proposed change.

Please verify that all attachments and references are relevant, current, and clearly labeled to avoid processing and review delays. Please list your attachments here:



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